PGR111, tasks for week 01

Task 1 – Your group

If you are going to work in a group: How can you use the newly established group to learn as much as possible in PGR111? Discuss and create some guidelines for how you can work well together. Feel free to talk about whether it might be appropriate to work together in the morning, until the start of classes.

Task 2 – Install MySQL

Install MySQL, as mentioned in the lecture and the "session 00" slides.

You will find information in the lecture material on Canvas about how to install, both for PC and Mac. But also ask fellow students or the teacher for help if you get stuck!

Important! Select a root password that you'll remember! (Maybe save it in a text file?)

Task 3 – Open and try out DBeaver (or MySQL Workbench)

Familiarize yourself with the DB manager tool:

1. Open **DBeaver** (or open and go to SQL Development in MySQL Workbench).
2. Try to set the **`world`** database as default. (Right click -> "Set as Default …".)
3. Click on the query window for the database.
4. Write a query that lists everything in the country table (**`SELECT * FROM Country`**), to check if querying works. (Click the icon to run the query or use the hotkey <CTRL+Enter> on windows and <CMD+Enter> on macOS).

Task 4 – Let's get started on SQL!

Write SQL queries that executes the following:

1. Retrieves all data from the **`City`** table.
2. Filters out all data for Norway (`CountryCode = 'NOR'`) in the **`City`** table.
3. Filters out city names and population (`Name, Population`) of cities in Norway in the **`City`** table.
4. Counts how many countries that are in the database. (Tip: See the functions on the slide at the end of the lecture.)

5. Sum the total population registered in the **`world`** database and calls the sum column for `WorldPopulation`.

6. Sum the total population in `Asia`.

7. Retrieve an ascending list over all countries and their total area, ordered by area.
   (Try to figure out (by yourself) which columns and table are needed.)

8. Makes a ranked list of countries by decreasing life expectancy and increasing population.

Task 5

Use the World database and think about what data it might be interesting to extract from it. Try to create some queries that give you this data. The queries must only use the SQL that we have reviewed so far (**`SELECT, FROM, WHERE, ORDER BY, AS, COUNT, SUM, AVG, MIN, MAX`**) and you need to fetch from one table.

Once you have successfully created a query that provides the expected answer, post the task in the Discussion in the Exercises block on Canvas. The above must describe the data to be retrieved. Feel free to mark the task with a degree of difficulty "easy", "medium" or "difficult" and say something about the expected result of the query. You are welcome to attach a hidden solution proposal by using the same color for text and background. If you cannot find the thread that will contain the week's student assignments, create it and give it a suitable name.

Example:

I want to get a sorted list (population) of all cities in Sweden. I'm trying my best, and I think I've found a good solution with:

```
SELECT * FROM City WHERE CountryCode = 'SWE' ORDER BY Population
DESC;
```

I can't find the thread with the week's student assignments, so I create one that I call: "Student-made assignments for week 01". As the first post, I enter my task and a hint about the correct result: "Medium: Get all cities in Sweden sorted by population (from largest to smallest). Expected result: 15 cities with Stockholm at the top."

Once you have created a few tasks and posted them in the forum, solve tasks in the forum at the level of difficulty that suits you.